The **33rd** Republic of China Biology Olympiad National Team Selection Camp , **2022**

Test Room **4**

Plant Group Practical Questions

Note: Operation time is limited, please make good use of the time waiting for the experiment to take effect.

ÿ Investigate the salt tolerance response of **ice plants** and observe leaf-related structures.

This practical exercise will treat ice plant with different salt concentrations to understand the effects of these different salt concentrations on ice plant growth.

The impact on growth.

I. Materials and Experimental Equipment:

A. Experimental plant materials:	quantity
1. For ice plant, on the 30th day of growth, begin watering with 0%, 0.1%, and 3% salt solutions respectively. Until the 40th day.	1 tray / 2 people

A. Experimental equipment:	Quantity A: Experimental equipment: 1 unit/person		quantity
1. Dissecting microscope and duplex microscope (Includes electronic eyepiece + laptop)	for each of the following: 11. Single-edged blade		1/person
2. P1000 micropipettes 3. 1000 µl	12. Glass slides/coverslips 1 box/person 13. Mortar		9 pairs/person
pipette tip 4. Oil-based ballpoint	and pestle		2 pairs/person
pen	1 person/support 14. Weighing plate		2 per person
5. Small scissors	1 tube/person 15. Colorimetric tube		2 pipes/person
6. Cloth gloves	1 pair/person 16.2 ml microcentrifuge tube		4 pipes/person
7. Timer	1 unit/person 17. Glass test tubes		2 pipes/person
8. Simple calculator 9.	One unit/person sharing the instrument:		
Tweezers (fine-tipped)	1 unit/person 1. Vacuum pump		Shared
10. Absorbent paper	1 box/person 2. Centrifuge		Shared
11. Petri dish (top + bottom)	1 group/person 3. Spectrophotometer		Shared
		4. Water bath (95 ° C)	Shared
		5. Electronic scale	Shared

B. Drug Solution: 1.	Quantity B of drug solution:		quantity
20% TCA containing 0.5% TBA (thiobarbituric acid)	10 ml 3. Liquid nitrogen		1 bottle
2. 5%(w/v)trichloroacetic acid (TCA)	4 ml	4. Distilled water	1 bottle

II. Experimental Methods, Results, and Discussion: Experimental

Methods: Seedlings of

Imperata cylindrica grown under white light (100 mmol/m² s) were transplanted into white light-treated soil. On day 30, seedlings were watered with 0%, 0.1%, and 3%

saline solutions, respectively, until day 40, which served as the observation period for this experiment.

Materials.

A. Measurement of MDA (Malondialdehyde) content in ice flower plants : MDA content can be used as a measure of oxidation resistance.

Environmental indicators.

1. Take 0.2 g of fresh *Ilex cornuta* leaf samples treated with 0% and 3% saline solutions respectively, grind them in liquid nitrogen, and then add 2 ml of 5% (w/v)

trichloroacetic acid (TCA) to form a homogenate. 2. Take 1 ml of the homogenate from the previous step (0% and 3%) and place them in two glass test tubes,

labeling them accordingly. 3. Take 4 ml of 20% TCA containing 0.5% TBA (thiobarbituric acid) and add it to the glass test tube from the previous step.

middle.

4. Place the glass test tube (from the previous step) into a 95°C container. the C. Heat on a test tube rack in a water bath for 30 minutes, then cover with a cloth.

Remove the suitcase from the water bath and let it cool for 2 minutes.

5. Transfer 2 ml of the mixture to microcentrifuge tubes (2 ml each), label them (exam number 0% and 3%), and centrifuge for 10 min (12000 xg). 6. Transfer the

supernatant to a new microcentrifuge tube, and

transfer approximately 1.5-1.7 ml of the supernatant to a colorimetric tube.

In the public area, the absorbance values of the blank and the OD532 and OD450 values of the sample were measured using a spectrophotometer.

Please note: When taking out the colorimetric tube, please take it from the top opening. The bottom is a light-transmitting area, so please do not touch it.

7. Data Paste Area:

8. Calculate the MDA content (mmol Lÿ¹) using the following formula:

Formula : **6.45 x OD532 – 0.56 x OD450.**

saline concentration (%)	MDA content (mmol Lÿ¹): MDA
Sample 10%	
Sample 2 3%	

B. Observation of the phenotypic morphology of the ice flower plant:

1. Take one mature leaf from each of the three salt concentrations (0%, 0.1%, and 3%) of *Impatiens balsamina* and observe them under a microscope.

Observe and record the morphology of sac-like cells on the leaf surface.

1-1. Three sampling points were randomly selected from the upper epidermis of each leaf, and the samples were fixed under a dissecting microscope.

Take magnified photographs to record the data, then count the number of cystic cells from the images. Triple coverage.

Experiment, calculate the average value, and fill it in the results and discussion section. referencePhotos must be archived for *jpg* File, set

The file name is the number of repetitions of the saline concentration, for example, the exam number. *: A1-0.1-1)*

1-2. Perform freehand sectioning using any leaf and prepare a water-embedded specimen, then place it under a duplex microscope.

Next, observe which part of the leaf the sac-like cells belong to: the hairs, the epidermis, or the mesophyll?

Take photos as evidence and explain the reasons for your judgment in the results and discussion section. Select the best photo

One file is archived *jpg* File, name the file a slice, for example, exam number. *: A1-* slice *)*

2. Take one mature leaf from each of the ice flowers cultivated at different salt concentrations, and cut two leaflets of appropriate size.

Place the leaf slices on glass slides, one on each side (front and back), add one or two drops of water, cover with another glass slide, and then...

Place them on a petri dish, with the front of the leaf facing up on the (seat number - top) petri dish and the back of the leaf facing down on the (seat number - bottom) petri dish.

The petri dish was then placed in the vacuum pump's shared area and evacuated for 5 minutes. The petri dish was then observed under a duplex microscope.

Guard cell structure: How to determine if stomata are open? Answer in the results and discussion section. Note

Select the best photo from each of the lower epidermis *jpg* File, named "Saline Concentration (Below)" Exam Number (Above

for archiving; for example, the pores. *: A1-0.1-* *)*

C. Results and Discussion:

1-1. Record the effects of saline concentrations (0%, 0.1%, and 3%) on the leaf growth of ice plant.

Are there many cystic cells distributed on the surface? Infer the possible functions of the cystic cells.

Answer: The number of sac-like cells on the epidermis of the leaves of the ice flower plant:

0%_______ÿ0.1%_______ÿ3%_______ (3 points each)

Possible functions: ____________________________ (5 points)

1-2. To which part of the leaf do these sac-like cells belong: the hairs, the epidermis, or the mesophyll?

(6 points: 3 points for location, 3 points for photo)

Answer:_____________________________

2. Distribution and opening/closing of stomata on the leaves of ice plants grown in saline solutions of different concentrations. (10 points each: 3 points for distribution, 3 points for opening/closing, 2 points each for upper/lower epidermal photographs; total 30 points)

Answer: 0 % saline solution: Stomatal distribution and opening/closing status: ___________(upper/lower; open/closed)

0.1% saline solution: Stomatal distribution and opening/closing status: ___________(upper/lower; open/closed)

3% saline solution: Stomatal distribution and opening/closing status: ___________(upper/lower; open/closed)

3. What is the MDA content of ice flower plants grown in 0% and 3% saline solutions? Explain the effect of saline concentration on the MDA content.

The effect of MDA content. (15 points)

Answer: MDA content of ice flower plants grown in 0 % saline solution: _________________

MDA content of ice flower plants grown in 3% saline solution: __________________

Relationship between saline concentration and MDA content: ___________________________________

4. Explain the effect of saline concentration on the form of carbon utilization during photosynthesis in ice plants. (10 points)

Answer:

5. Does the amount of MDA affect the photosynthetic process of ice flower plants? Explain. (10 points)

Answer:

Yes: ___________. No: ___________.

reason:

6. Based on this experiment, compare the differences in photosynthetic modes between C3, C4, and CAM type plants.

No. Please compare the efficiency of stomatal opening and closing time, carbon utilization, and water utilization (6 points). What type of carbon utilization does the ice flower belong to? (3 points) Explain (6 points).